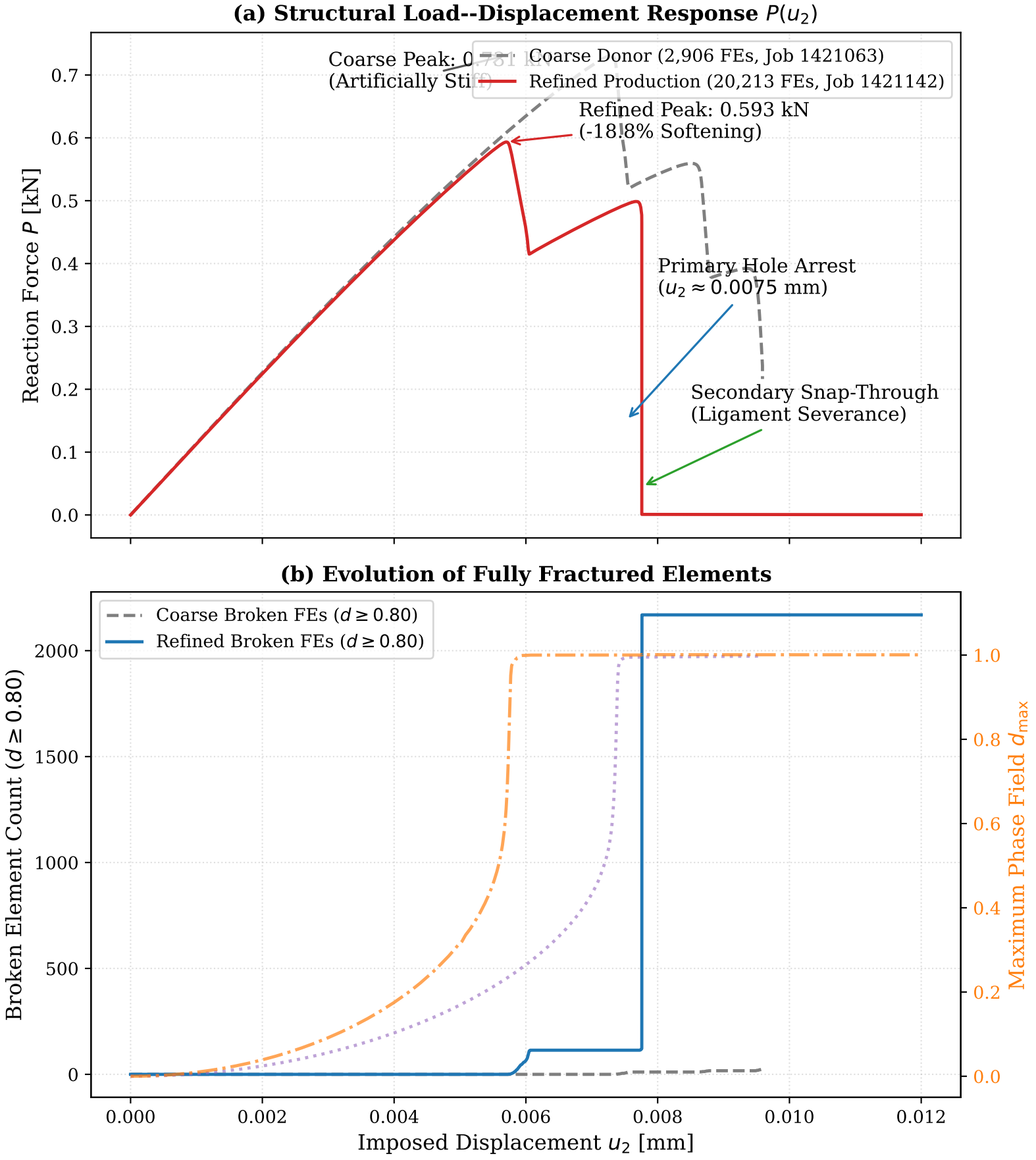
(a) Structural Load--Displacement Response $P(u_2)$
Coarse Donor (2,906 FEs, Job 1421063)
Refined Production (20,213 FEs, Job 1421142)
Refined Peak: 0.593 kN
(-18.8% Softening)
Primary Hole Arrest
($u_2 \approx 0.0075$ mm)
Secondary Snap-Through
(Ligament Severance)
Reaction Force $P$ [kN]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
(b) Evolution of Fully Fractured Elements
Coarse Broken FEs ($d \geq 0.80$)
Refined Broken FEs ($d \geq 0.80$)
Broken Element Count ($d \geq 0.80$)
0
500
1000
1500
2000
Maximum Phase Field $d_{max}$
0.0
0.2
0.4
0.6
0.8
1.0
0.000
0.002
0.004
0.006
0.008
0.010
0.012
Imposed Displacement $u_2$ [mm]